

TITANIC DATA CLEANING AND PREPROCESSING

Synent Technologies Data Science Internship

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Objective

The objective of this project is to clean and preprocess the Titanic dataset by handling missing values, removing duplicates, converting data types, and renaming columns.

Dataset Information

Dataset Name: Titanic Dataset

Source: Kaggle

Rows: 891

Columns: 12

Methodology

The following preprocessing steps were performed:

- Loaded dataset using Pandas
- Examined data structure and missing values
- Filled missing values in Age using median
- Filled missing values in Embarked using mode
- Removed Cabin column due to excessive missing values
- Checked and removed duplicate records
- Converted data types where necessary
- Renamed columns for better readability
- Results

The dataset was successfully cleaned.

Missing values were handled appropriately.

The cleaned dataset was exported as cleaned_titanic.csv.

Conclusion

Data preprocessing is an important step in the data science workflow. After cleaning, the Titanic dataset became suitable for further analysis and machine learning tasks.